

Deconstructing the Mad Genius: An Analysis of the Neuroscientific Relationship between Depression and Creativity

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Abstract

This paper explores the connection between depression and creativity, particularly as it relates to the character of the “mad genius”. It delves into various neuroscientific models for depression and depressive symptoms and attempts to illustrate a link with “generative creativity.” The goal of this paper is to connect two topics that are generally studied separately with the goal of destigmatizing mental health or isolating creative people experiencing mental illness. The eventual conclusion is that there is evidence that the default mode network (DMN) is hyperactive and the central executive network (CEN) hypoactive in depressed individuals, and the same is true for people excelling in generative creativity. The practical implication of this is there is a correlation between the neural correlates of depression and the neural correlates required for creation of different literary devices, including verse, poetry and lyrics. The paper also explores how several works from the authors Sylvia Plath and Alice Walker can be used as case studies to illustrate the aforementioned connections between depression and creativity.

Introduction

This section has been significantly abridged from the original paper, and only the most important topics have been included. This is far from an exhaustive review of existing literature on the topic and the conversation on mental health is long and ongoing. With that said, the histories of mental illness and creativity are currently and have always been inextricably intertwined, most notably in the ancient Greek idea that creatives had been blessed by divine Muses, allowing them to see truths beyond the boundaries of logic. The intersection of these topics has never been comprehensively explored, but what began as a casual interest in people seemingly blessed with divine foresight has evolved into a token character:

a creative person who is able to see connections between disparate parts of the world, and they find it so isolating that they are driven to madness. A “mad genius.”

Sensationalization of the mentally ill artist functions to dehumanize and ultimately exacerbate feelings of alienation and isolation, both for the artist and their observers. “This reductive view of our patients can be dehumanizing and will prevent the clinician from understanding their patient as a whole,” which is a barrier both to healthcare professionals and loved ones of those with mental illness. The process of othering a mentally ill artist feeds heavily into the sense of isolation that is already pervasive in many psychiatric disorders, including Major Depressive Disorder (MDD), which is the primary focus of this paper.

Depression is an incredibly heterogeneous condition, meaning that it manifests variably from person to person. Symptoms can include anything from a classic depressed mood, to somatic symptoms like appetite alterations, to feverish bursts of energy that can almost never be sustained. “Depression is a disorder of mood, so mysteriously painful and elusive in the way it becomes known to the self ... it thus remains nearly incomprehensible to those who have not experienced it in its extreme mode,” writes William Styron, illustrating that the study of depression in an objective sense is nearly impossible, even to those who have had their own experiences with the illness. As such, further exacerbating existing depressive feelings and symptoms is harmful to those experiencing depression, even without the added pressure of creative production.

With this said, the relationship between creativity and mental illness is undeniable and this paper is an attempt at holistically viewing the two in terms of their neural correlates and literary manifestations. The observations and conclusions in this review will be informed by neurophysiological frameworks as well as

the lived experiences of individuals with depression. Following a neuroscientific breakdown of the “mad genius” model, art and literary works can be used to foster empathy and understanding, promoting connection rather than isolation and support rather than the glorification of suffering. This is reflected in the literary works of writers like Sylvia Plath and Alice Walker, whose works can be used to illustrate how mental state informs the content and structure of creative literature.

At its core a “mad genius” is a person experiencing mental illness that manifests as a pain that is nearly impossible to articulate or convey. Thus, this person turns to creativity – metaphor, allegory and imagery become tools of expression. When sensationalization is removed, “explorations of depression in literature and art are full of metaphor and visual cues that promote a deeper understanding, which can ultimately facilitate our capacity for compassionate care.”

The Neuroscience of Depression...

MDD’s heterogeneity has been well established and there is a documented lack of consensus on how to categorize depression. Do we categorize by symptom? By duration? Do we even need to divide it up? The most universally accepted classification of depression today is articulated in the DSM-5, in which various cognitive and somatic symptoms including anhedonia, difficulty communicating, sense of worthlessness or guilt, increased or decreased sleep and “depressed mood” are listed. Attempts at classifying depression through the last century have varied considerably. There was an initial attempt at differentiating between “endogenous” and “chemical” depression, followed by a revolutionary focus on neurotransmitters as facilitators of depression and a more recent exploration into the structure and function of the depressed brain.

As a result of the scientific gridlock that’s resulted from attempts at categorizing depression, the vast majority of research into the neural correlates and treatment for depression focuses on the symptoms, rather than the root cause or broader neural networks at play in the illness. There is a significant body of knowledge as to which brain structures influence anhedonia or impair decision making, and how SSRIs serve to facilitate the alleviation of depressive symptoms. While this was a logical pivot based on the need to treat depression, it is limiting because it makes conclusive generalizations risky, even when significant research has been conducted. Additionally, it prevents us from studying depression alongside its

many comorbidities, including anxiety and ADHD. Nevertheless, some neural correlates for depressive symptomatology have been identified and more research is conducted daily as the pressing need for treatment of MDD continues.

Perhaps the most well-known correlate of depression is hyperactivity within the amygdala, a structure with immense influence over emotional response and regulation. Excessively sensitive response to negative stimuli such as sad or fearful faces has been noted in depressive patients in comparison to controls, and amygdalic hyperactivation has also been shown to correlate with excessive rumination, the persistent, passive focus on negative stimuli. Furthermore, patients with depression show greater blood-oxygen level dependent (BOLD) response to negative stimuli in areas involved in response to salient emotional information, including the amygdala, insula and ACC. This results in an individual being particularly sensitive to negative input, explaining the negative bias that is characteristic of depression. This negative bias is also in large part due to reduced functional connectivity in the dorsolateral prefrontal cortex (dlPFC), which renders it less able to provide top-down regulation of the limbic system. The coupling of the hyperactive limbic system’s bias towards the negative and reduced BOLD response in the dlPFC results in an individual that is hypersensitive to negative stimuli and entirely unable to escape the ruminative spiral that follows.

Another generally well-established neural correlate of depression is hippocampal hypoactivity, which has been proven to not affect the encoding of new memories, but rather the retrieval of existing memories. This has been further proven to cause long-term consequences in patients with chronic episodic depression. Namely, reduced hippocampal volume has been observed in patients experiencing extended depressive episodes over the course of years. Additionally, functional connectivity studies have shown decreased connections between the CA1 neurons projecting from the hippocampal heads to the medial prefrontal cortex (mPFC) and decreased fractional anisotropy (FA) between the hippocampus and frontal regions. The overall effect of this is an impaired ability to retrieve existing memories and an inability to regulate both decision-making and self-referential thought.

Finally, there are various frontal structures implicated in depressive symptoms, including decreased grey matter in left middle frontal gyrus, which is associated with emotional bias, apathy and

loss of motivation. There is also cataloged dysfunction in many of the circuits involving the prefrontal cortex, including the prefrontal-subcortical cortex, which contributes significantly to dysregulation of executive function. Finally, many people experiencing depression exhibit significant dysfunction along the fronto-striatal axis and a general decrease in grey matter intensity in the ventral striatum, which is associated with habit formation and reward-based learning. This is associated with the suicidality that many patients with depression experience and, in conjunction with the other listed symptoms, results in creating an entirely inhospitable mental state. Understanding these symptoms has been critical in prioritizing and targeting the most harmful symptoms of depression with treatments like SSRIs, different therapy approaches and even more experimental treatments like psychedelics. The question remains, however, of how each of these symptoms and structures fit together and if it's possible to rationalize and articulate the lived experience of depression from these data.

...and Creativity, and Why It Matters That We Study Them Together

The recent development of the Network Theory is an interesting framework that's slowly gained traction in the last 20 years as a mechanism for studying neural activity. The general idea is that the brain can be divided into networks, defined based on what function they serve in the larger environment of the brain. Very broadly, the Default Mode Network (DMN) is active when the brain is at rest, responsible for internally oriented and self-regulating thought processes. It generally includes the medial prefrontal cortex (mPFC), the posterior cingulate cortex (PCC), precuneus, hippocampus and parahippocampal cortex, with extensive connectivity to the amygdala. The Central Executive Network (CEN) is thought to support higher order cognitive processes like decision making and goal-directed actions, and is generally accepted to include the dlPFC, the anterior cingulate cortex (ACC) and inferior parietal lobe. The Salience Network (SN), including the insula and dorsal ACC, serves as a regulating body and works to filter important stimuli and initiate transitions between the DMN and CEN.

This theory of network connectivity, rather than explaining affective processes like decision-making or emotional processing, functions to categorize based on overall brain state and level of processing. Its relevance to depression, although not a perfect model, is better than many that have come before. The CEN's higher-order cognitive processes are

significantly impaired in individuals with MDD, with decreased functional connectivity observed on fMRIs as well as decreased white matter integrity observed on FA. This correlates with previous studies regarding the dlPFC and its reduced top-down control of the limbic system (see above). This means that more effort is required to perform higher-order processing tasks like problem solving, sustained attention and goal-directed behaviors. As a result, the DMN is observed to become increasingly active in order to compensate for dysfunction in the CEN. Hyperactivity in the DMN has been linked to symptoms like excessive rumination, focus on negative thoughts and self-referential negativity. Additionally, due to its significant functional connectivity to the amygdala, DMN hyperactivity has downstream effects on amygdalic upregulation, contributing to negative bias, anhedonia and derealization. Taken together, these data indicate that depression can be modeled by hyperactivity in the DMN that results from reduced activity in the CEN. An interesting note that supports this theory is the fact that experienced meditators (meditation is a proven treatment for depression) exhibit more heavily regulated DMN activity. It's possible that the reduction in depressive symptoms that results from meditation could be attributed to downregulation of a patient's hyperactive DMN, further supporting the network model for depression.

What is particularly interesting about network theory is that it has also been proven to be a generally reliable mechanism of categorizing creativity as well. In her 2018 book *The Neuroscience of Creativity*, Dr. Anna Abraham identifies two main types of creativity: the first is generative creativity (also called open or divergent creativity), which is categorized as being open-ended and exploratory. Generative creativity relies on a widened attentional capacity and ability to make broad and seemingly disassociated connections. The second is evaluative creativity (also called closed or convergent creativity), which is rooted in interpretation and analysis, rather than generation.

She continues to establish that engagement of the DMN is associated with generative creativity, particularly the precuneus and hippocampus, which are involved in episodic memory retrieval, self-related operations, visuo-spatial imagery and flexible cognition. Spontaneous emotional creativity is the result of hyperactivity in the DMN and engagement of emotional processing structures. It is this creativity that facilitates story generation, which Abraham states engages the inferior frontal gyrus, as well as identifying positive correlations between verbal divergent thinking

and white matter integrity in the ACC, as well as grey matter in the dlPFC and striatum.

Downregulation of the CEN leads to reduced cognitive control, which allows for more loose associations, which are the random connections artists occasionally identify between seemingly unrelated (but usually poetic) things. Individuals experiencing a downregulated CEN also experience disinhibition when it comes to goal-directed tasks, a widened attentional capacity and a broader scope of creative techniques accessible. This transient hypofrontality has been shown to result in a lapse of the explicit or conscious system, allowing for the DMN to take over in a “seamless sensory-motor integration process.” In other words, an artistic flow state.

Finally, although it is generally accepted as part of the DMN, recent studies have found that it is also analytically feasible to decouple the DMN and the hippocampus in the case of depression. Hippocampal reactivation for memory retrieval is broadcast via midline hubs including the mPFC, which is notably downregulated in depressed patients, resulting in the hippocampal degradation noted. This would pose a challenge to the network model – how can the DMN be generally upregulated in depressed patients while the hippocampus is not only downregulated, but actively experiencing volume reduction? This is where the decoupling potential comes in: there is a paradoxical increase in “spontaneous thought,” in depressed patients, which is thought to be related to functional connectivity between the hippocampus and the broad DMN. Thus, while the DMN is hyperactive, the functional connectivity between its elements is reduced, resulting in depression’s effects on the hippocampus.

Thus, there is significant evidence for a connection between the correlates of depression and creativity. The parallels don’t end at the network level, however. Similar parallels exist at the circuitous and structural levels as well. For example, when the DMN is hyperactive, we’ve established that an individual experiences depressive symptoms generally, as well as an increase in spontaneous emotional creativity. On a more structural level, the inferior frontal gyrus experiences higher fractional amplitude of low-frequency fluctuations (fALFF) to compensate for the decreased grey matter caused by DMN upregulation. The inferior frontal gyrus is associated with language production and cognitive control and typically exhibits increased activity in individuals with depression. It is also the only structure that is truly involved in all

elements of literary creativity as defined by Abraham: processing metaphors, brainstorming and the creation of prose and poetry.

Additionally, DMN upregulation features activation of the precuneus, which is associated with increased divergent thinking and creativity, as well as decreased activation in poetic interpretation, biasing an individual towards creation rather than consumption and analysis. Activation of the precuneus in conjunction with the decreased functional connectivity characteristic of depression results in relatively unregulated self-referential thought, contributing to depressive rumination. Finally, although not actually part of the DMN, the strong functional connectivity the network has with both the striatum and the amygdala means that they are impacted by alterations to DMN regulation. This is particularly apparent because a decrease in grey matter in the striatum results in paradoxical executive dysfunction and verbal divergent thinking. Finally, the dlPFC, which is a critical part of the CEN, also plays a role in the depressive creative experience. Responsible for the production of pleasurable emotions, depressed patients exhibiting downregulated dlPFC exhibited a desire but not an ability to produce positive emotions. This, coupled with the hypoactive dlPFC’s allowance for looser creative composition and tendency towards metaphor, makes poetry a very convenient medium for expression.

Thus, it becomes clear that network analysis reveals several parallels between the depressed and creative mental states. The downregulation of the CEN and higher-order processing allows for the domination of a freer, less streamlined state. For better and for worse, this state, the upregulation of the DMN, results in both depressive symptoms like rumination, negative bias and obsessive self-referential thought, and an elevated capacity for many literary devices, like metaphor, personification and vivid imagery. This knowledge can be extrapolated in an application to the actual works of authors with mental illness, exploring both what is said and how it is said. The final goal of this article is to establish a more holistic framework through which we can use literary devices and themes as mechanisms of understanding and empathizing with the parts of the depressive experience that can’t be explicitly articulated. Recognizing these creative devices as tools of expression and connection, rather than some kind of inexplicable divine foresight, functions to break the stigma surrounding depression and provide support for those in need.

Depression and Creativity in the Works of Sylvia Plath & Alice Walker

The section is abridged from the initial paper written for this project, which featured separate sections for each author with the intention of analyzing their individual styles and circumstances. For the purpose of this article, the sections have been combined and the most important themes have been highlighted, but it is not a comprehensive analysis of this topic. With that said, the first theme I've identified in Plath's and Walker's writing is a tension between their reality and what they want. In other words, the desire for a pleasurable feeling or better world and an inability to create it or believe in it.

On the personal scale, Plath tackles this idea in her poem, "Years," in which she identifies her intense repulsion with the idea of stagnancy. "Eternity bores me," she writes, "I never wanted it. / What I love is / the piston in motion – / my soul dies before it. / And the hooves of the horses, / their merciless churn." Here, Plath utilizes imagery and poetic cadence to mimic the sound and feel of horses, galloping endlessly and freely, to convey her fear of stasis. She radiates a desire to be unmoored and free, which is also reflected in her resistance to traditional gender roles. "...I never wanted to get married," she writes in *The Bell Jar*, her only novel published before her death. "The last thing I wanted was infinite security and to be the place the arrow shoots off from. I wanted change and excitement and to shoot off in all directions myself, like the colored arrows from a Fourth of July rocket." The novel is a fictional rendition of Plath's own coming of age, in which she struggles to rationalize the many challenges that come with being twenty and depressed. This line speaks to Plath's own feelings of being held down or held back by the weight of societal expectations and her own struggles with mental health.

This tension between the present reality and hope for the future is brought to a large scale in Walker's writing. Alice Walker's life was characterized by activism and ferocious engagement with the world around her. She was immensely invested in the movements for civil rights and women's rights and the happenings of her surroundings heavily impacted her mental health. "The history / of this assault is long / and so is memory / among the poor," she writes, breaking up her lines in an effort to force the reader to slow down, to feel the memory. It is interesting that she associates memory here with a salient emotion, because coupling memory with a feeling often makes it more easily retrievable in the long run.

She is worn down by the conflict between her

reality and the future she envisions for herself and those she cares for. "The isolation. The racism. The sexism. The slow breakdown of my spirit after I'd finished this novel or that, this story or that, this poem or that, and looked about and found little to amuse, divert, or sustain me." Here, Walker contextualizes her feelings living as a Black woman in the South during the Civil Rights movement. Once again, her sentences are short and meaningful. The cadence of her writing parallels the debilitating patterns that eventually wear her down and her final line speaks to another symptom and theme in the writing of individuals experiencing depression: anhedonia.

Alice Walker frequently returns to the idea of death. She has a casual relationship with it, frequently contemplating the idea of suicide with a nonchalant interest, even as "in my emotional life, I feel I am stagnating ... Nothing is missing from this success except my capacity any longer to feel it." Taken from an entry in her personal journal following the rousing success of *The Color Purple*, this line articulates the anhedonic feelings of apathy and isolation Walker experienced. The reader gets a sense that Walker felt unable to detach from her inner mind and connect with the outside world, as if she was watching all of her success on a movie screen instead of experiencing it. This sense of detachment and apathy is a central theme in Plath's *The Bell Jar* as well, with the central metaphor circling around the idea that "...wherever I sat...I would be sitting under the same glass bell jar, stewing in my own sour air." This central metaphor encapsulates the story and after reading it for the first time, the reader begins to recognize it in many of Plath's descriptions of apathy, detachment and hopelessness. It demonstrates the rumination she was experiencing, as well as her feelings of isolation. Later in the book, Plath describes recovery as the bell jar being lifted, reconnecting her to the world. But she also conveys the fear that, at any moment, the bell jar may descend again, shutting her off and isolating her in her sour air once more.

Herein lies another theme in depression writing. In her 1962 poem "Elm," Plath writes, "I am inhabited by a cry. / nightly it flaps out / looking, with its hooks, for something to love. / I am terrified by this dark thing / that sleeps in me; / all day I feel its soft feathery turnings, its malignity." In these lines, Plath personifies her mind, giving it wings and anger and the ability to drag her down at a moment's notice. Her use of metaphor here speaks to the internalization of her illness; she has immense turmoil within her mind and struggles desperately to externalize it in any

meaningful way. These lines also speak to the cyclical nature of depression and the threat that it could return at any moment. For Plath, it is a terrifying and isolating feeling.

The counterpart of this internalization can often be seen in Walker's writings, however. Walker is enthralled by the natural world and her love for it spills over into her writing. Celie, her protagonist in *The Color Purple*, does not have an easy life and experiences significant abuse and hardship at the hands of her father, husband and the society in which she lives. She is anhedonic – she moves through life woodenly, going through the motions without joy, anger or love for herself or the world. The epistolary structure of the book allows the reader to watch Celie find her voice and her letters reflect her journey with her own self-image, her relationships and her belief system. Shug eventually tells her, "God love all them feelings. That's some of the best stuff God did ... I think it pisses God off if you walk by the color purple in a field somewhere and don't notice it." Walker uses imagery to illustrate the need for externalization, the need to disrupt negative rumination. Celie's story is about so much more than mental illness, but this passage serves as a simple reminder of how all-encompassing anhedonia and detachment can feel, and the need to resist the feelings of isolation and internalization that hard times foster.

One thing about depression is that it comes in waves. When a person is in the trough of the wave, in a depressive episode, it's nearly impossible to remember or to believe that there will be another crest in the future. It's difficult not to feel crushed by the endless spiral of pain and apathy. "To be warm, to be dry / to be writing poems again / (after months of distraction / and emptiness!), / to love and be loved / in absentia / is joy enough for me," writes Walker, illustrating a simple need that she and Plath share. "I want to write because I have the urge to excel in one medium of translation and expression of life. I can't be satisfied with the colossal job of merely living. Oh, no, I must order life in sonnets and sestinas and provide a verbal reflector for my 60-watt lighted head," writes Plath in her personal journal. For Plath and Walker, writing is a need. It is an expression that isn't matched by speech, or movement, or art. Writing was more than an outlet, but an act of resistance against a world that made them feel isolated, helpless and hopeless, in one way or another.

Finally, and perhaps most importantly, depression is a disease that radically alters self-image. It seems to sleep, and slowly steals a person's ability to feel,

then their ability to connect, and finally, when it seems there is nothing left to take, depression shatters the sense of self. For Plath and Walker, writing was the lifeline keeping them tethered to their sense of self. "Before I leave the stage / I will sing the only song / I was meant truly to sing. / It is the song / of I AM / Yes: I am Me / & / You. / WE ARE." Walker's passage feels like an homage to what she had overcome to reach the publication of this book, a mantra of self-determination that was earned, forcefully. Writing became not only an emotional outlet, but an affirmation of existence, a grounding in the external. Writing down a mantra solidifies it – in a mind that feels untrustworthy, or untrue to the self, having words on a page reclaims agency and externalizes mental turmoil. Thus, when Sylvia Plath wrote, "I took a deep breath and listened to the old brag of my heart, she probably did just that. "I am, I am, I am."

Conclusion

There is no definitive conclusion on the neural roots of depression. In a similar vein, there are an infinite number of ways to categorize creativity. There is no possible way to fully synthesize the correlations between creativity and depression in a comprehensive sense, and yet they are two states that are so deeply intertwined. Studying them together allows for the deconstruction of a harmful character and is a step towards a world where healthcare is a little bit less dehumanizing. The connection between mental illness and creativity is not magic – it is tangible, chemical and theoretically, understandable.

There is power in connection. Depression is an illness that exacerbates the feeling of isolation, even between people experiencing the same thing. In connecting these seemingly disparate ideas, we open the door for unification instead of isolation. "We encounter among others / walking there / a sense of what we have lost: / the broad expanse of humanity's / sensitivity to the oneness of itself." There is a loss of this sense of unity in the broader academic and intellectual communities. Many topics are disconnected when, in a practical sense, nearly every field has some combination of science, humanities and the arts. There is progress to be made in destigmatizing both mental health and emotional expression and the deconstruction of the "mad genius" monolith is a good place to start.

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